

Phantom: Distributional Forecasting for Cross-Sectional Cryptocurrency Return Prediction

Alexandre Dalban

CentraleSupélec, Université Paris-Saclay

Contact: alexandre.dalban@student-cs.fr, alexandre.dalban@gmail.com

Abstract

Cross-sectional return prediction—ranking which assets will outperform their peers—is a cornerstone of quantitative investing, yet remains underexplored in cryptocurrency markets where distributional properties (heavy tails, regime shifts) demand more than point forecasts. We present Phantom, a Transformer-based model that predicts full Student- t distributions of cross-sectional *relative* returns across 362 cryptocurrencies at horizons of 1–30 days, using only standard OHLCV price and volume features. The model employs a patched encoder with cross-attention decoding for simultaneous multi-horizon prediction, trained on cross-sectionally demeaned returns to isolate idiosyncratic signal from shared market factors. On 15 months of out-of-sample data (January 2025–March 2026), drawn from a survivor-biased universe, Phantom achieves a rank information coefficient (IC) of 0.124 at the 10-day horizon ($t = 8.93$, Newey-West; conservative overlap-adjusted lags still give $t \approx 6-7$), with a gross, horizon-corrected long-short Sharpe ratio of approximately 4.1 (an upper bound; see Section ??), a 74.7% win rate over overlapping 10-day holding windows, and near-perfect decile monotonicity (Spearman $\rho = 0.988$; quintile $\rho = 1.000$). Double-sorted portfolio analysis reveals that Phantom’s signal is substantially correlated with the low-volatility anomaly ($\rho = 0.63$)—a naive low-volatility sort attains a higher raw IC (0.165)—but Phantom adds incremental ranking power within mid- and high-volatility terciles, and combining both signals yields the highest IC of 0.157. Through systematic ablation across eight model versions, we establish that the cross-sectional signal resides in daily OHLCV patterns, cannot be extracted from absolute returns, does not benefit from derivative market features (funding rates, order flow), and scales with the breadth of the asset universe—but not with higher-frequency data.

Keywords

cryptocurrency, cross-sectional prediction, distributional forecasting, Transformer, Student- t

1 Introduction

The cryptocurrency market, with a combined capitalization exceeding \$2 trillion and continuous 24/7 trading across hundreds of liquid assets, presents a distinctive opportunity for cross-sectional return prediction. Unlike directional forecasting—predicting whether an asset’s price will rise or fall—cross-sectional prediction concerns the simpler but economically powerful task of *ranking*: identifying which assets will outperform or underperform their peers over a given horizon. Recent work has established that cross-sectional factors familiar from equity markets, including momentum [1], size, and volume, carry significant premia in cryptocurrency [2].

However, existing approaches rely exclusively on point forecasts, discarding the distributional information that is critical for position sizing, risk management, and tail-risk awareness in a market characterized by heavy tails, extreme volatility clustering, and frequent regime shifts.

This paper addresses a gap at the intersection of distributional forecasting and cross-sectional prediction. While probabilistic time series models such as DeepAR and Temporal Fusion Transformers produce predictive distributions for individual assets, they are not designed for cross-sectional ranking. Conversely, cross-sectional factor models [1, 2] produce rankings but provide no uncertainty quantification. Distributional foundation models such as JointFM [5] have shown that Transformer architectures can recover complex distributional structure from time series, but have only been evaluated on synthetic data with no demonstrated cross-sectional signal on real financial assets. We bridge these approaches.

We present **Phantom**, a Transformer-based model that predicts calibrated Student- t distributions of cross-sectional *relative* returns across 362 cryptocurrencies at horizons of 1–30 days simultaneously. The architecture consists of a patched Transformer encoder that processes 120 days of 6-channel OHLCV features, a cross-attention decoder that queries all 30 horizons in a single forward pass, and a Student- t output head with 3 parameters per horizon (location μ , scale σ , degrees of freedom ν). The model is trained on cross-sectionally demeaned returns—each asset’s return minus the same-day cross-sectional mean—which removes shared market factors and isolates the idiosyncratic signal that drives relative performance.

On 15 months of out-of-sample data (January 2025 to March 2026, 430 trading days) from a survivor-biased universe, Phantom achieves a rank information coefficient (IC) of 0.124 at the 10-day horizon (Newey-West $t = 8.93$; conservative overlap-adjusted lags give $t \approx 6-7$), increasing to 0.166 at 30 days. A long-short portfolio that buys the top quintile and shorts the bottom quintile earns a gross, horizon-corrected Sharpe ratio of approximately 4.1—an upper bound given the survivor-biased universe and the absence of spread, impact, and borrow costs (Section ??)—with a 74.7% win rate over overlapping 10-day holding windows, quintile return monotonicity of 1.000 (decile 0.988), and IC positive in all 15 out-of-sample months. The strategy’s breakeven transaction cost is approximately 34 basis points one-way, modest headroom above Binance’s standard 10 bps fee that would be largely consumed by realistic spread and impact costs.

These results emerge from a systematic search over the design space, spanning eight model versions that progressively identify where cross-sectional signal resides—and where it does not.

Contributions.

- We propose Phantom, the first distributional forecasting model for cross-sectional cryptocurrency return prediction,

producing calibrated Student- t distributions at 30 horizons simultaneously.

- Through systematic ablation across eight model versions, we establish that (i) cross-sectional signal exists in daily OHLCV features but only in *relative* (not absolute) returns, consistent with weak-form market efficiency; (ii) synthetic SDE pretraining achieves oracle accuracy on synthetic data but transfers zero signal to real assets; (iii) derivative market features (funding rates, taker buy ratios) provide no incremental IC; and (iv) higher-frequency (4-hour) data degrades performance due to sample overlap, establishing that the signal is daily-scale.
- We demonstrate that long-short performance improves with asset universe breadth (60 to 362 assets) through a mechanical diversification effect—mean IC actually declines slightly (0.14 to 0.12)—and that the gross signal exceeds exchange-fee-level costs (breakeven ≈ 34 basis points one-way), though realistic spread, impact, and borrow costs would consume much of it.
- We characterize the signal’s relationship to the low-volatility anomaly via double-sorted portfolios: a naive low-vol sort attains a higher raw IC (0.165 vs. 0.124), but Phantom’s incremental predictive power concentrates in mid- and high-volatility assets (spread $t=2.6$ and 4.1) where low-vol sorting alone provides no ranking, and combining both signals achieves the highest IC of 0.157.

2 Related Work

2.1 Cross-Sectional Cryptocurrency Factors

Liu, Tsyvinski, and Wu [1] establish a three-factor model for cryptocurrency returns comprising market (CMKT), size (CSMB), and momentum (CMOM) factors, constructed analogously to the Fama-French framework for equities. Momentum emerges as the dominant cross-sectional predictor, with weekly long-short returns and t -statistics well above 3, substantially exceeding the momentum premium observed in equity markets. Their work demonstrates that standard asset pricing methodology transfers to the crypto asset class, though the factor zoo is far smaller than in equities. Fieberg et al. [2] extend this line of work by constructing CTREND, a composite trend factor that aggregates momentum and moving-average crossover signals across multiple lookback horizons. CTREND subsumes any individual momentum measure and earns significant alpha over the three-factor model, demonstrating that multi-horizon trend aggregation is more informative than any single-horizon signal.

The low-volatility anomaly (low-volatility assets earn higher risk-adjusted returns than high-volatility ones) is well-documented in equities [9] and has been observed in cryptocurrency markets, where the retail-dominated investor base and lottery-like preferences for high-volatility tokens provide a natural explanation. Our double-sort analysis (Section 5.4) reveals substantial interaction between our model’s signal and the low-volatility effect.

All existing cross-sectional crypto studies produce point forecasts: they rank assets by a scalar signal and evaluate via portfolio sorts. None provides distributional predictions that quantify the

uncertainty of the ranking, which is essential for position sizing and risk management.

2.2 Transformer-Based Time Series Forecasting

Nie et al. [3] introduce PatchTST, which segments time series into subseries-level patches before applying a Transformer encoder, reducing sequence length while capturing local semantic information. Their finding that channel-independent processing outperforms channel mixing on standard benchmarks provides a foundation for our patched encoder design, though we depart from channel independence by jointly embedding all six OHLCV features within each patch, as cross-channel interactions (e.g., between returns and volume) are central to our prediction task. Ansari et al. [4] propose Chronos, a time series foundation model that tokenizes real-valued observations into discrete bins and trains a T5-based language model on a large corpus of diverse time series, enabling zero-shot probabilistic forecasting across domains. While both PatchTST and Chronos advance time series forecasting, neither is designed for cross-sectional ranking or the heavy-tailed distributional properties specific to financial returns.

2.3 Distributional Foundation Models

Most closely related to our work, Hackmann [5] proposes JointFM, a foundation model for multivariate distributional time series forecasting. JointFM uses Variable-Factored Attention to model both temporal and cross-variable dependencies and employs parametric distributional heads—multivariate Gaussian mixtures and multivariate Skewed Student- t distributions—to produce full joint predictive distributions over multiple coupled targets. Trained on synthetic data generated from stochastic differential equations (GBM, Merton jump-diffusion, Bates stochastic volatility, regime-switching), JointFM achieves strong distributional calibration as measured by energy distance [8]. However, JointFM remains evaluated *exclusively* on synthetic processes and does not address real financial data, leaving open the question of whether synthetic pretraining transfers to actual asset returns. Our work answers this question directly: through systematic comparison of synthetic and real-data pretraining (Section 5.7), we find that synthetic SDE pretraining achieves oracle distributional accuracy on synthetic test data but transfers zero cross-sectional signal to real cryptocurrency markets. Phantom extends the distributional forecasting paradigm to real cross-sectional prediction with demonstrated out-of-sample signal.

2.4 Probabilistic Forecasting

Distributional forecasting of financial returns has a long history, from GARCH-family models to deep learning approaches including DeepAR and Temporal Fusion Transformers. We evaluate distributional quality using the Continuous Ranked Probability Score (CRPS), a strictly proper scoring rule [6]:

$$\text{CRPS}(F, y) = \mathbb{E}_F[|X - y|] - \frac{1}{2} \mathbb{E}_F[|X - X'|] \quad (1)$$

where X, X' are independent draws from the predictive distribution F and y is the realized value. Strict propriety ensures that the score is uniquely minimized when the forecaster reports the true conditional distribution, providing a principled training and evaluation objective.

We parameterize predictive distributions as Student- $t(\mu, \sigma, \nu)$ rather than Gaussian for two reasons: (i) cryptocurrency returns exhibit substantial excess kurtosis even at multi-day horizons, and (ii) the degrees-of-freedom parameter ν provides an adaptive, learnable measure of tail heaviness per asset and horizon. Our finding that the model consistently pushes $\nu \rightarrow 2$ (the minimum for finite variance) confirms that heavy-tailed modeling is essential for this asset class and that Gaussian assumptions are inadequate.

3 Methodology

3.1 Problem Formulation

Let $\mathcal{A}_t = \{1, \dots, N_t\}$ denote the set of actively traded cryptocurrency assets on date t . For each asset i , we observe a context window of daily OHLCV (open, high, low, close, volume) data spanning $L = 120$ trading days, from which we compute a feature matrix $\mathbf{x}_i \in \mathbb{R}^{L \times C}$ with $C = 6$ channels (Section 3.2). We define the *cumulative log return* of asset i over the next h days as

$$r_{i,t,h} = \sum_{k=1}^h \log\left(\frac{c_{i,t+k}}{c_{i,t+k-1}}\right) \quad (2)$$

where $c_{i,t}$ is the closing price. To isolate idiosyncratic signal from shared market movements, we work with *relative returns* (Section 3.5):

$$r_{i,t,h}^{\text{rel}} = r_{i,t,h} - \frac{1}{N_t} \sum_{j \in \mathcal{A}_t} r_{j,t,h} \quad (3)$$

The model f_θ produces a parametric predictive distribution for each asset and horizon simultaneously:

$$f_\theta(\mathbf{x}_i) = \{(\mu_{i,h}, \sigma_{i,h}, \nu_{i,h})\}_{h=1}^H, \quad H = 30 \quad (4)$$

where each triplet parameterizes a Student- t distribution: $r_{i,t,h}^{\text{rel}} \sim \text{Student-}t(\mu_{i,h}, \sigma_{i,h}, \nu_{i,h})$. The model serves a dual objective: (1) *calibrated distributional forecasts*—the predicted distributions should match the empirical frequency of outcomes—and (2) *cross-sectional ranking signal*—on each date, the predicted location parameters $\{\mu_{i,h}\}_{i \in \mathcal{A}_t}$ should rank assets in an order correlated with their actual relative performance.

3.2 Feature Engineering

From raw OHLCV data $(o_t, h_t, l_t, c_t, v_t)$ for each asset, we compute six channels that are universal across all tradeable assets:

Table 1: Input feature channels.

Ch	Feature	Formula
0	Log return	$\log(c_t/c_{t-1})$
1	Intraday range	$(h_t - l_t)/c_t$
2	Body ratio	$(c_t - o_t)/(h_t - l_t)$
3	Log volume ratio	$\log(v_t/\tilde{v}_{30})$
4	Trailing volatility	$\hat{\sigma}_{30} \cdot \sqrt{252}$
5	Momentum	$\sum_{k=0}^9 r_{t-k}$

Here $\tilde{v}_{30}$ is the 30-day rolling median volume and $\hat{\sigma}_{30}$ is the 30-day rolling standard deviation of log returns, annualized by $\sqrt{252}$. Channels 1–2 capture intraday price dynamics (volatility and candle structure), channels 3–4 capture volume and realized volatility,

and channel 5 captures short-term trend. Critically, we apply *no instance normalization* (e.g., RevIN [10]) or mean-absolute scaling, as our ablation studies (Section 5.7) found that such normalizations destroy the cross-sectional signal by removing the relative magnitude differences between assets that the model exploits for ranking.

3.3 Model Architecture

Patched encoder. The input $\mathbf{x}_i \in \mathbb{R}^{L \times C}$ is divided into $N = L/P$ non-overlapping patches of length $P = 5$ time steps, giving $N = 24$ patches. Each patch $\mathbf{p}_n \in \mathbb{R}^{P \times C}$ is flattened and linearly projected into $\mathbb{R}^d$:

$$\mathbf{z}_n^{(0)} = \mathbf{W}_{\text{proj}} \cdot \text{flatten}(\mathbf{p}_n) + \mathbf{e}_n \quad (5)$$

where $\mathbf{W}_{\text{proj}} \in \mathbb{R}^{d \times PC}$ is the projection matrix and $\mathbf{e}_n \in \mathbb{R}^d$ is a learned positional encoding, with $d = 512$. The resulting sequence $\mathbf{Z}^{(0)} \in \mathbb{R}^{N \times d}$ is processed by an M -layer pre-norm Transformer encoder ($M = 8$, GELU activation, $d_{\text{ff}} = 2048$, 8 attention heads), producing contextualized patch representations $\mathbf{Z}^{(M)} \in \mathbb{R}^{N \times d}$. The full architecture is shown in Figure 1.

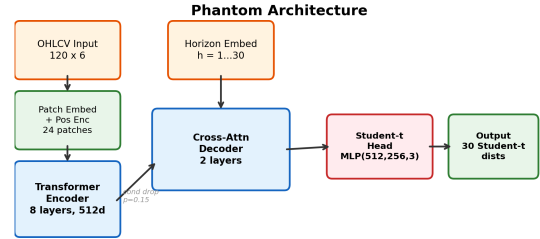


Figure 1: Phantom architecture. The patched encoder processes 120 days of 6-channel OHLCV input into 24 contextualized patch tokens. The cross-attention decoder simultaneously queries all 30 horizons against the encoder output. Each decoded vector is mapped to Student- t parameters via a two-layer MLP.

Cross-attention decoder. To decode distributions at all $H = 30$ horizons in a single forward pass, each horizon $h \in \{1, \dots, H\}$ is mapped to a query $\mathbf{q}_h^{(0)} = \mathbf{E}_h$ via a learned embedding table. The 30 queries attend to the encoder output through $D = 2$ decoder layers, each applying pre-norm cross-attention followed by a feedforward network:

$$\begin{aligned} \hat{\mathbf{q}}_h^{(t)} &= \mathbf{q}_h^{(t-1)} + \text{MultiHead}\left(\underbrace{\text{LN}(\mathbf{q}_h^{(t-1)})}_{\text{query}}, \underbrace{\mathbf{Z}^{(M)}}_{\text{key, value}}\right) \\ \mathbf{q}_h^{(t)} &= \hat{\mathbf{q}}_h^{(t)} + \text{FFN}(\text{LN}(\hat{\mathbf{q}}_h^{(t)})) \end{aligned} \quad (6)$$

This produces a decoded vector $\mathbf{q}_h^{(D)} \in \mathbb{R}^d$ per horizon, with total complexity $O(H \cdot N \cdot d)$.

Student-t output head. The decoded vector $\mathbf{q}_h^{(D)} \in \mathbb{R}^d$ for each horizon is mapped to distribution parameters via a two-layer MLP:

$$\begin{aligned} (\hat{\mu}, \log \hat{\sigma}, \log \hat{\nu}) &= \mathbf{W}_2 \text{GELU}(\mathbf{W}_1 \mathbf{q}_h^{(D)} + \mathbf{b}_1) + \mathbf{b}_2 \\ \sigma_h &= \exp(\text{clamp}(\log \hat{\sigma}, -7, 2)) \\ \nu_h &= \exp(\text{clamp}(\log \hat{\nu}, \log 2.01, \log 100)) \end{aligned} \quad (7)$$

The clamping ensures $\sigma_h \in [e^{-7}, e^2]$ and $\nu_h \in [2.01, 100]$. The lower bound $\nu > 2$ guarantees finite variance; our ablation finds the model consistently pushes $\nu \approx 2$, indicating maximally heavy-tailed distributions.

Condition dropout. During training, the encoder output is zeroed with probability $p_{\text{drop}} = 0.15$:

$$\tilde{\mathbf{z}}^{(M)} = \mathbf{Z}^{(M)} \cdot \mathbf{m}, \quad m_i \sim \text{Bernoulli}(1 - p_{\text{drop}}) \quad (8)$$

applied per sample (not per token). This trains the decoder to produce reasonable predictions even without encoder input, preventing the decoder from learning to ignore the encoder via its residual connection—an analog of classifier-free guidance in diffusion models.

3.4 Training Objective

The loss function combines four terms:

$$\mathcal{L} = \lambda_{\text{NLL}} \mathcal{L}_{\text{NLL}} + \lambda_{\text{CRPS}} \mathcal{L}_{\text{CRPS}} + \lambda_{\text{MSE}} \mathcal{L}_{\text{MSE}} + \lambda_{\text{var}} \mathcal{L}_{\text{var}} \quad (9)$$

with weights $\lambda_{\text{NLL}} = 1.0$, $\lambda_{\text{CRPS}} = 0.5$, $\lambda_{\text{MSE}} = 0.3$, $\lambda_{\text{var}} = 0.1$.

Negative log-likelihood. The primary loss is the NLL of the observed relative return under the predicted Student- t distribution, averaged across all horizons and samples:

$$\mathcal{L}_{\text{NLL}} = -\frac{1}{BH} \sum_{b,h} \log p(r_{b,h}^{\text{rel}} | \mu_{b,h}, \sigma_{b,h}, \nu_{b,h}) \quad (10)$$

where $p(\cdot | \mu, \sigma, \nu)$ is the Student- t density.

CRPS. The CRPS term (Eq. 1) is evaluated on a subset of horizons $\{1, 5, 10, 15, 20, 25, 30\}$ days using the sample-based representation with 256 Monte Carlo draws from the predicted distribution.

Mean MSE with horizon weighting. To encourage non-trivial conditional means (preventing collapse to $\mu = 0$), we add a weighted MSE loss on the predicted mean:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{B} \sum_b \sum_{h=h_{\min}}^H w_h (\mu_{b,h} - r_{b,h}^{\text{rel}})^2, \quad w_h = \sqrt{h/H} \quad (11)$$

The $\sqrt{h}$ weighting reflects the signal-to-noise ratio of drift: for a random walk with drift δ , $\text{SNR}(h) = \delta\sqrt{h}/\sigma \propto \sqrt{h}$. We set $h_{\min} = 5$ to avoid penalizing mean predictions at very short horizons where SNR is negligible.

Encoder variance penalty. To prevent representation collapse (all encoder outputs becoming identical regardless of input), we add a penalty on the variance of encoder representations across the batch: $\mathcal{L}_{\text{var}} = -\text{Var}_b[\tilde{\mathbf{z}}_b]$ where $\tilde{\mathbf{z}}_b$ is the mean-pooled encoder output for sample b .

3.5 Cross-Sectional Demeaning

The relative return (Eq. 3) removes the shared market factor from each asset's return on each date. If all assets rise 5% but asset i rises 8%, its relative return is +3%; an asset rising only 2% receives −3%. This is analogous to the market-adjustment step in Fama-MacBeth regressions and serves two purposes: (1) it makes the prediction task *market-neutral*, so the model need not predict market direction, and (2) it isolates the *idiosyncratic* component of returns that drives cross-sectional dispersion.

The demeaning is computed offline over all (t, h) pairs before training. We require a minimum of 3 assets per date; 99.9% of samples are adjusted. The demeaned target has mean approximately zero by construction, but its *cross-sectional variance* varies substantially across dates and horizons—this variance is what the model's $\sigma_{i,h}$ learns to predict.

4 Experimental Setup

4.1 Data

We collect daily OHLCV data for all actively traded USDT-denominated pairs on Binance. After filtering stablecoins, leveraged tokens, and assets with fewer than 300 days of history, our universe contains **362 cryptocurrencies** spanning August 2017 to March 2026. Because the universe is defined by trading status at data-collection time (March 2026), pairs delisted at any earlier point—including during the test window—are excluded retroactively; the universe is therefore survivor-biased, and all reported results should be read as upper bounds (Section ??).

Table 2: Dataset statistics.

	Train	Validation	Test
Period	< 2024-01	2024	$\geq$ 2025-01
Samples	229,759	96,966	144,630
Unique dates	2,208	366	430
Assets per date	$\sim$ 160	$\sim$ 290	$\sim$ 340

The temporal split ensures out-of-sample evaluation: the validation year (2024) acts as a buffer between training and test data. Windows adjacent to the split boundaries share up to 30 days of context or target overlap; a purge gap of the maximum horizon length between splits would remove this residual leakage (Section ??). The model is initialized from a checkpoint pretrained on a broader universe of 413 assets (including equities, forex, and commodities) before fine-tuning on the crypto-only dataset.

4.2 Baselines

We compare against the following cross-sectional strategies, all evaluated using the same long-short portfolio methodology (top/bottom quintile, equal-weighted, daily rebalancing):

- **Momentum (kd):** rank assets by cumulative log return over the past $k \in \{5, 10, 20\}$ days. This is the dominant factor in the crypto literature [1].
- **Mean-reversion (kd):** rank by *negative* past $k \in \{1, 5\}$ day return.

- **Volume:** rank by mean log volume ratio (channel 3) over the past 5 days.
- **Low volatility:** rank by *negative* trailing realized volatility (channel 4). The low-volatility anomaly is well-documented across asset classes [9].
- **Random:** Gaussian noise signal (null hypothesis).

4.3 Evaluation Metrics

Cross-sectional signal. The primary metric is the *rank information coefficient* (IC): the Spearman rank correlation between predicted $\mu_{i,h}$ and realized $r_{i,h}^{\text{rel}}$, computed per date and averaged. We report Newey-West adjusted *t*-statistics [7] throughout to account for autocorrelation in daily IC and return series. The *long-short portfolio* buys the top quintile (~ 72 assets) and shorts the bottom quintile by predicted μ , equal-weighted, rebalanced daily. We report annualized Sharpe ratios using a 365-day convention (crypto trades 24/7). Because the long-short series at horizon h consists of h -day forward returns sampled daily—overlapping observations—naive daily annualization with $\sqrt{365}$ would inflate the Sharpe ratio by approximately $\sqrt{h}$; all Sharpe ratios in this paper are therefore *horizon-corrected* by dividing the naive figure by $\sqrt{h}$ (Section ??).

Transaction costs. We compute net returns under one-way cost assumptions of 0, 5, 10, 20, 30, and 50 basis points, applied proportionally to *turnover*—the fraction of portfolio positions that change each day. We report the *breakeven cost*: the maximum one-way fee at which the strategy’s mean daily return reaches zero.

Distributional quality. We evaluate calibration via the probability integral transform (PIT): if the predicted distribution F_θ is correctly calibrated, the PIT values $F_\theta(r^{\text{rel}})$ should be uniformly distributed. We report coverage at the 50%, 80%, 90%, and 95% levels and the Kolmogorov-Smirnov test statistic for PIT uniformity.

5 Results

All results in this section are on the out-of-sample test set (January 2025 to March 2026, 430 trading days, 144,630 samples across 362 assets); recall that the universe is survivor-biased (Section 4.1), so all figures are upper bounds.

5.1 Cross-Sectional Signal

Figure 2(a) shows the rank IC by forecast horizon. IC increases monotonically from 0.080 at the 1-day horizon to 0.166 at 30 days, consistent with the theoretical scaling of drift signal-to-noise ratio with $\sqrt{h}$. At the reference 10-day horizon, the mean IC is 0.124 with a Newey-West *t*-statistic of 8.93, and IC is positive on 78% of trading days and in all 15 out-of-sample months.

5.2 Portfolio Performance

Table 3 presents the quintile portfolio sort at the 10-day horizon. Returns increase monotonically from Q1 (−1.49% per 10-day window) to Q5 (+1.04%), yielding a long-short spread of 2.53% per 10-day holding window with a Newey-West *t*-statistic of 7.44 (Andrews-rule lags; conservative overlap-adjusted lags reduce this, Section ??). The Spearman rank correlation between quintile rank and mean return is 1.000; the finer decile sort is near-perfectly monotonic (0.988).

Table 3: Quintile portfolio sort (h=10d). Newey-West *t*-statistics in parentheses. Returns are per overlapping 10-day holding window; Sharpe ratios are gross and horizon-corrected (Section ??).

	Q1	Q2	Q3	Q4	Q5	Q5−Q1
Return (%/10d)	−1.49 (−7.71)	−0.24 (−1.80)	0.18 (1.83)	0.48 (4.64)	1.04 (5.05)	2.53 (7.44)
Sharpe	−4.2	−0.9	0.8	2.2	2.8	4.1

Figure 2(b) shows the cumulative overlapping long-short spreads under different cost scenarios. The gross, horizon-corrected annualized Sharpe is approximately 4.1, with an annualized gross return of approximately 93%; both are upper bounds given the survivor-biased universe. After 10 bps of one-way costs the net Sharpe falls to roughly 2.9, and at 30 bps the strategy operates close to its breakeven one-way cost of approximately 34 bps—above Binance’s standard 10 bps fee, but with no allowance for spread, market impact, or borrow costs (Section ??). The gross Sharpe reflects both the model’s IC and the diversification benefit of trading 362 assets: for a long-short strategy, Sharpe scales approximately as $\text{IC} \times \sqrt{N}$ where N is the number of assets. Average daily quintile-membership turnover is 37.8%.

Table 4: Strategy performance summary (h=10d, top/bottom 20%, equal-weighted). All figures gross unless noted, horizon-corrected, and computed on a survivor-biased universe (upper bounds; Section ??). Win rate is over overlapping 10-day holding windows; max drawdown is measured on the overlapping spread series and is indicative only.

Metric	Value	Metric	Value
Ann. return (gross)	~93%	Win rate (10d win.)	74.7%
Ann. volatility	~23%	Max drawdown	−38.6%
Sharpe (gross)	~4.1	Avg. turnover	37.8%
Sharpe (10 bps)	~2.9	Breakeven cost	~34 bps
Sharpe (30 bps)	~0.4	IC (NW <i>t</i>)	0.124 (8.93)
Quintile monot.	1.000	L/S ret (NW <i>t</i>)	7.44
Decile monot.	0.988	IC > 0 days	78%

5.3 Baseline Comparison

Table 5 compares Phantom to simple cross-sectional strategies. All momentum signals have *negative* IC during the test period, indicating a mean-reverting regime in the 2025–2026 crypto market—consistent with the literature finding that crypto momentum is time-varying [2]. Mean-reversion signals are weakly positive but not statistically significant. The strongest baseline is the **low-volatility** strategy (IC = 0.165), which we analyze further in Section 5.4.

5.4 Relationship to the Low-Volatility Anomaly

The low-volatility baseline achieves IC = 0.165, raising the question of whether Phantom simply rediscovers this known anomaly. We

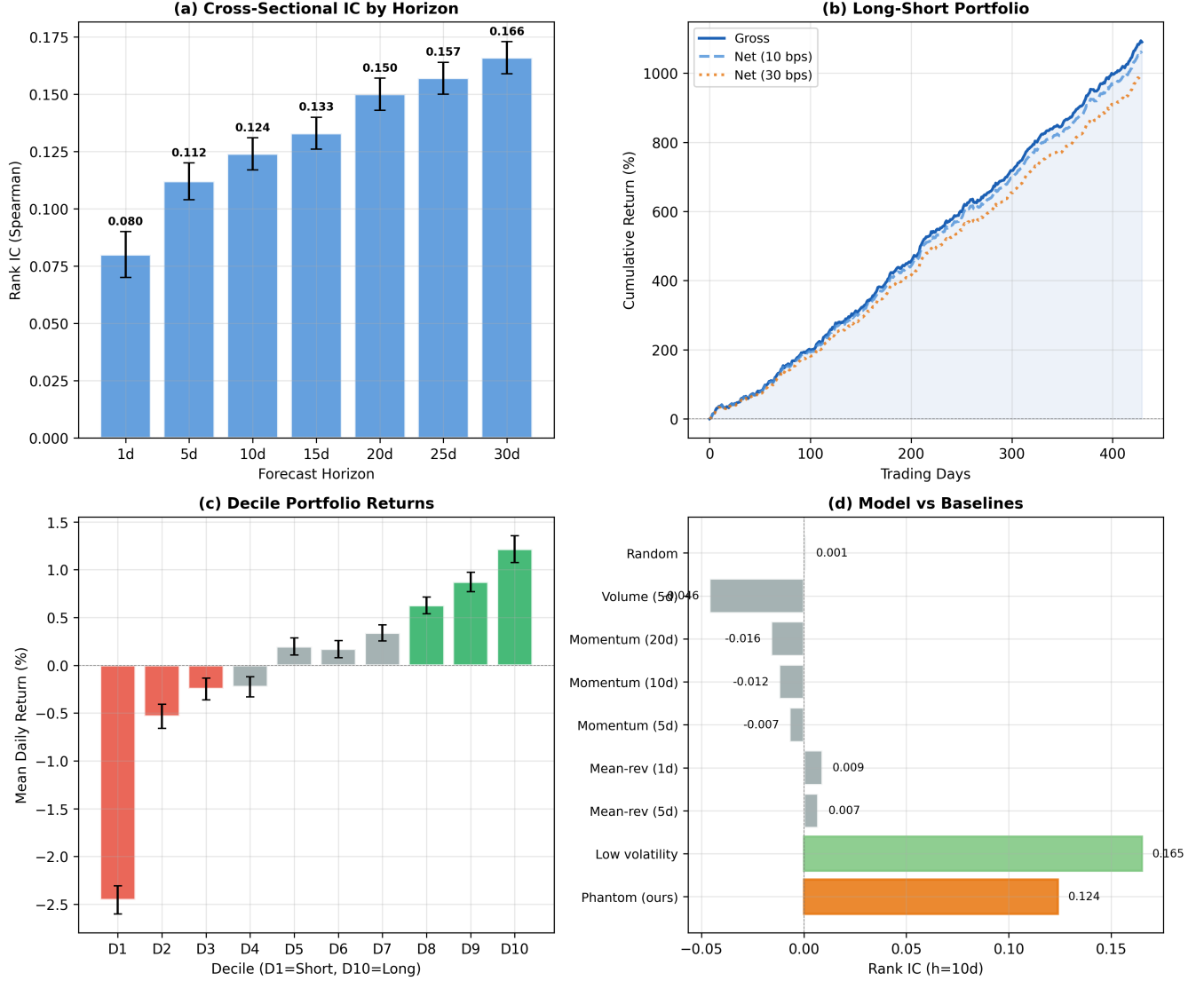


Figure 2: Main results. (a) Rank IC increases with horizon, consistent with drift $\text{SNR} \propto \sqrt{h}$ (longer horizons also overlap more heavily). (b) Cumulative overlapping 10-day long-short spreads under different cost assumptions—indicative of signal strength, not an implementable equity curve. (c) Decile portfolio returns are near-perfectly monotonic (Spearman 0.988). (d) Phantom attains the highest IC among directional baselines; the naive low-volatility sort achieves a higher raw IC (Section 5.4).

investigate via double-sorted portfolios: on each date, we independently sort assets into volatility terciles and Phantom prediction terciles, then examine returns within each cell (Figure 3a).

The cross-sectional correlation between Phantom’s signal and the low-vol signal is $\rho = 0.63$, indicating substantial overlap. However, Phantom’s ranking power *concentrates in the high-volatility tercile* (IC = 0.092, NW $t = 6.77$), precisely where sorting by volatility alone provides no differentiation—all high-vol assets earn negative relative returns, but Phantom distinguishes the “less bad” from the “worst.” Combining both signals (equal-weighted rank average) yields the highest IC of 0.157, exceeding either signal alone.

5.5 Distributional Quality

Beyond ranking, the model produces calibrated uncertainty estimates (Figure 4). The PIT histogram is approximately uniform, and coverage calibration is excellent: 90% prediction intervals contain 90.3% of realized returns. The predicted volatility term structure follows $\sigma_h \propto \sqrt{h}$ at short horizons before diverging upward at longer horizons, reflecting the excess volatility clustering in crypto returns. The model consistently learns $\nu \approx 2$, the minimum degrees of freedom for finite variance, confirming that cryptocurrency relative returns have substantially heavier tails than a Gaussian model would accommodate.

Table 5: Baseline comparison (h=10d). Sharpe ratios are gross and horizon-corrected; Win% is over overlapping 10-day windows.

Strategy	Rank IC	Sharpe	Win%
Phantom (ours)	0.124	4.1	74.7
Momentum (5d)	-0.007	-0.3	49.1
Momentum (10d)	-0.012	-0.3	45.8
Momentum (20d)	-0.016	-0.3	48.6
Mean-rev (1d)	0.009	0.3	50.0
Mean-rev (5d)	0.007	0.3	50.9
Volume (5d)	-0.046	-2.8	33.7
Low volatility	0.165	3.8	73.3
Random	0.001	0.1	53.3

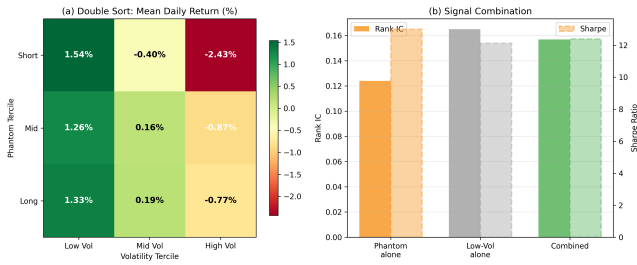


Figure 3: (a) Double sort: mean daily returns by volatility and Phantom tercile. The volatility column dominates (low-vol assets earn positive returns regardless of Phantom ranking), but Phantom’s long-short spread is significant within mid-vol ($t=2.6$) and high-vol ($t=4.1$) terciles. (b) Combining both signals yields the highest IC.

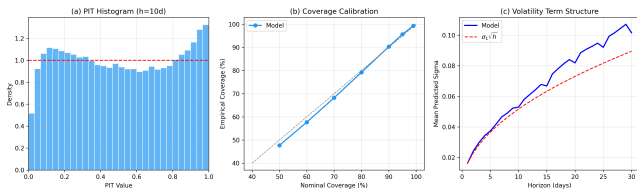


Figure 4: Distributional quality. (a) PIT histogram (h=10d) is approximately uniform. (b) Coverage calibration closely tracks the diagonal. (c) Predicted σ follows $\sqrt{h}$ scaling at short horizons.

5.6 Robustness

Figure 5 demonstrates the temporal stability of the signal. The rank IC is positive in all 15 out-of-sample months, ranging from 0.037 (March 2026, partial month) to 0.200 (November 2025). The 60-day rolling horizon-corrected Sharpe ratio is positive 100% of the time, with a minimum of approximately 0.9 and a mean of approximately 4.4. Daily quintile-membership turnover is stable around 38%, with no regime-dependent spikes. The maximum drawdown on the overlapping spread series is -38.6% with a 14-day recovery, occurring during a broad crypto deleveraging event.

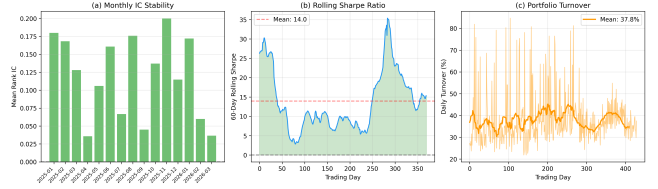


Figure 5: Robustness. (a) Monthly IC is positive in all 15 months. (b) 60-day rolling horizon-corrected Sharpe always positive (min ≈ 0.9). (c) Turnover stable at 38%.

5.7 Ablation: What Works and What Doesn’t

Table 6 summarizes the progression across eight model versions, each testing a specific design hypothesis. The key inflection point is the shift from absolute to relative returns ($v4 \rightarrow v5$), which unlocks cross-sectional signal that was inaccessible in all prior versions. Widening the asset universe from 60 to 362 assets ($v6 \rightarrow v8$) substantially improves the long-short Sharpe ratio without any architectural change; this is a mechanical diversification effect of breadth—mean IC actually declines slightly ($0.14 \rightarrow 0.12$)—rather than a stronger per-asset signal. Every negative result—synthetic pretraining, derivative features, and sub-daily data—eliminates a plausible design choice. Figure 6 shows the two-stage training dynamics for the final model.

5.8 Limitations

Five limitations qualify the headline results; all reported performance figures should be read as upper bounds.

Survivorship bias. The universe comprises all Binance USDT pairs with trading status active at data-collection time (March 2026). Pairs delisted at any point during 2017–2026—including during the test window—are excluded retroactively from training, testing, and the cross-sectional mean that defines the relative-return target. Because delistings concentrate among poorly performing assets, both the IC and the long-short spread are biased upward, and the demeaning target itself is contaminated. Reconstructing a point-in-time universe is the single most important robustness step for future work.

Overlapping-return annualization. The long-short series consists of 10-day forward returns sampled daily. An earlier version of this analysis annualized this series with $\sqrt{365}$ as if the observations were daily returns, inflating the Sharpe ratio by approximately $\sqrt{10}$ (yielding reported values of 11.8–13.0), the annualized return (925%), and the breakeven cost (336 bps), and mislabeling the 74.7% win rate of overlapping 10-day windows as a daily win rate. We state this error openly: all figures in the present version are horizon-corrected (gross Sharpe ≈ 4.1 , annualized gross return $\approx 93\%$, breakeven ≈ 34 bps one-way), and the correction changes the economic interpretation from implausibly large to strong-but-plausible for a gross, survivor-biased backtest.

Implementability and costs. The cost model applies proportional fees to quintile-membership turnover only; bid-ask spread, market impact, and borrow costs are not modeled, and both legs equal-weight illiquid tail names. Much of the bottom quintile has no

Table 6: Ablation across model versions (OOS crypto). “–” = not applicable. Sharpe ratios are gross and horizon-corrected (Section ??); earlier drafts reported these values inflated by $\sim\sqrt{10}$ through naive daily annualization of overlapping 10-day returns.

Ver	Change	IC	Sharpe	Finding
v1	Synthetic pretrain	–	–	Oracle CRPS on synthetic
v2	+ Student- t head	–	–	Same CRPS, simpler
v3	Real multi-asset	0.00	–	Good calibration, no signal
v4	Multi-horizon abs.	0.00	–	Memorizes, doesn't transfer
v5	Relative returns	0.09	1.5	Signal unlocked
v6	+ Funding/taker buy	0.14	1.7	Features don't help
v7	4h bars	0.10	0.8	Worse: sample overlap
v8	362 assets	0.12	4.1	Sharpe gains via breadth

margin or perpetual-futures market and is therefore not shortable in practice. Accounting for these frictions, we consider a realistic net-of-cost Sharpe ratio in the range of 1–2 plausible, rather than the gross upper bound of ≈ 4.1 .

Overlap with the low-volatility effect. The signal is rank-correlated 0.63 with inverse trailing volatility, and a naive low-volatility sort attains a higher raw IC (0.165 vs. 0.124). Phantom’s incremental contribution is statistically significant only within the mid- and high-volatility terciles (spread $t = 2.6$ and 4.1); claims of dominance over all baselines are not supported, and the appropriate reading is that Phantom adds signal *conditional* on volatility rather than replacing the low-vol factor.

Overlap-adjusted inference and split hygiene. The daily IC and spread series at the 10-day horizon carry an MA(9) dependence from overlapping targets; the Andrews-rule Newey-West lag of 5 under-corrects for it. With more conservative lag choices the IC t -statistic falls from 8.93 to roughly 6–7—still significant, but the sample contains only ~ 43 independent 10-day periods. In addition, windows adjacent to the train/validation/test boundaries share up to 30 days of context or target; inserting a purge gap of the maximum horizon length between splits would eliminate this residual overlap and is recommended for future revisions.

6 Conclusion

We presented Phantom, a Transformer-based distributional forecasting model for cross-sectional cryptocurrency return prediction. By training on cross-sectionally demeaned (relative) returns across 362 cryptocurrencies, the model achieves a rank IC of 0.124 (Newey-West $t = 8.93$; roughly 6–7 under conservative overlap-adjusted lags) with a gross, horizon-corrected long-short Sharpe ratio of approximately 4.1—an upper bound on a survivor-biased universe—a 74.7% win rate over overlapping 10-day holding windows, and near-perfect decile monotonicity (0.988; quintile 1.000) on 15 months of out-of-sample data.

The systematic ablation across eight model versions yields several insights: (i) absolute returns are unpredictable from OHLCV, but cross-sectional *relative* returns contain exploitable signal; (ii) synthetic SDE pretraining achieves oracle accuracy but transfers zero signal to real markets; (iii) derivative features add no value over

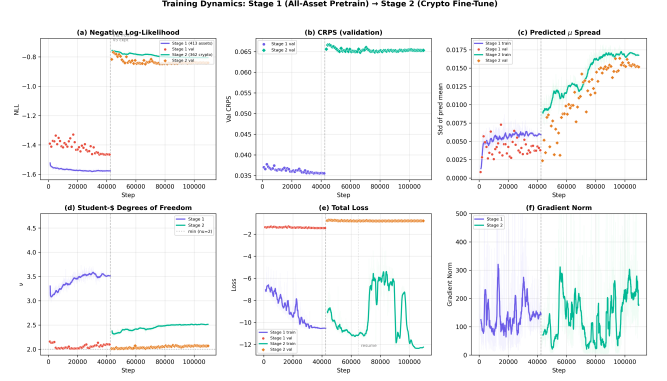


Figure 6: Two-stage training dynamics. Stage 1 (purple): pre-training on 413 multi-asset samples. Stage 2 (teal): fine-tuning on 362 crypto assets (dashed line = init from Stage 1 checkpoint). (a) NLL improves in both stages. (b) Val CRPS drops after crypto fine-tuning. (c) Predicted μ spread grows in Stage 2. (d) v stabilizes near 2. (e) Total loss with resume point. (f) Gradient norm increases during adaptation.

price and volume; and (iv) signal scales with universe breadth but not with data frequency.

Limitations. Section ?? details the main caveats: the universe is survivor-biased, the headline figures are gross of realistic frictions (a net-of-cost Sharpe of 1–2 is more plausible than the gross ≈ 4.1), and inference must account for overlapping 10-day returns. In addition, the signal is crypto-specific—equities show near-zero IC—our evaluation does not include live trading validation with real execution, the signal is substantially correlated with the low-volatility anomaly ($\rho = 0.63$), and the distributional output (σ , v) is not yet exploited for adaptive position sizing.

Future work. The most important next steps are methodological: re-evaluation on a point-in-time universe that includes delisted pairs, and retraining with a purge gap of the maximum horizon length ($H = 30$ days) between the train/validation/test splits. Further directions include liquidity-screened, cost-aware portfolio construction, incorporation of alternative data (on-chain metrics, social sentiment), adaptive position sizing using the predicted $\sigma_{i,h}$, and extension to decentralized finance (DeFi) protocols where cross-sectional prediction over liquidity pools presents similar opportunities.

Reproducibility. All code, model checkpoints, and processed datasets are publicly available at <https://github.com/ADnocap/phantom>.

References

- [1] Y. Liu, A. Tsyvinski, and X. Wu. Common risk factors in cryptocurrency. *Journal of Finance*, 77(2):1133–1177, 2022.
- [2] C. Fieberg, T. Hornuf, G. Lau, and A. Mihailova. A trend factor for the cross section of cryptocurrency returns. *Journal of Financial and Quantitative Analysis*, 2024.
- [3] Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam. A time series is worth 64 words: Long-term forecasting with transformers. *ICLR*, 2023.
- [4] A. F. Ansari, L. Stella, et al. Chronos: Learning the language of time series. *arXiv:2403.07815*, 2024.

- [5] D. Hackmann. JointFM: A joint foundation model for distributional prediction. *arXiv:2603.20266*, 2026.
- [6] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.
- [7] W. K. Newey and K. D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987.
- [8] G. J. Székely and M. L. Rizzo. Energy statistics: A class of statistics based on distances. *Journal of Statistical Planning and Inference*, 143(8):1249–1272, 2013.
- [9] A. Ang, R. J. Hodrick, Y. Xing, and X. Zhang. The cross-section of volatility and expected returns. *Journal of Finance*, 61(1):259–299, 2006.
- [10] T. Kim, J. Kim, Y. Tae, C. Park, J. Choi, and J. Choo. Reversible instance normalization for accurate time-series forecasting against distribution shift. *ICLR*, 2022.
- [11] I. Aldasoro, S. Barth, and M. Tsomocos. Crypto carry. *BIS Working Paper 1087*, 2024.

A Model Hyperparameters

Table 7: Full model configuration for v8 (final model).

Hyperparameter	Value
<i>Architecture</i>	
Context length L	120 days
Input channels C	6 (OHLCV)
Patch length P	5
Number of patches N	24
Model dimension d	512
FFN dimension d_{ff}	2048
Encoder layers M	8
Decoder layers D	2
Attention heads	8
Forecast horizons H	30
Head hidden dimension	256
Total parameters	31.7M
<i>Training</i>	
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$)
Peak learning rate	3×10^{-4}
Min learning rate	1×10^{-6}
LR schedule	Cosine annealing
Warmup steps	2,000
Weight decay	0.05
Gradient clipping	1.0
Batch size	512
Dropout	0.3
Condition dropout p_{drop}	0.15
AMP	bfloat16
<i>Loss weights</i>	
λ_{NLL}	1.0
λ_{CRPS}	0.5
λ_{MSE}	0.3
λ_{var}	0.1
Min MSE horizon h_{min}	5 days
Horizon weighting	$\sqrt{h/H}$
<i>Output head</i>	
σ range	$[e^{-7}, e^2]$
v range	$[2.01, 100]$